Jonathan Dambros

Porto Alegre, Brazil · jonathanwvd@gmail.com · GitHub · LinkedIn · Portfolio · Google Scholar

PROFILE

Engineering Manager & Researcher with expertise in agentic systems, LLM evaluation, and industrial AI. Ph.D. in Chemical Engineering with 10+ years in Python, process monitoring, and optimization. Experienced leading cross-functional teams and delivering production-grade AI frameworks with measurable impact.

EXPERIENCE

Engineering Manager — LLM Evaluation, Turing 2023 – Present

- Lead a team of 30+ reviewers evaluating LLM outputs with calibrated rubrics.
- Designed evaluation frameworks and reviewer workflows for reliability and consistency.
- Managed multiple projects ensuring model quality and robustness.

Postdoctoral Researcher, UFRGS — GIMSCOP 2019 – Present

- Developed Python libraries (DigitalTwin, Kalman) in collaboration with Petrobras.
- Researched and deployed monitoring and diagnostics systems in process industries.
- Published peer-reviewed work on control-loop performance monitoring and optimization.

Team Lead (Self-employed), Latos 2020 – 2023

- Built data-driven solutions for manufacturing and process industries.
- Developed and deployed "Heron," a web platform for data science workflows.
- Integrated multi-source pipelines with visualization in Grafana, Power BI, Tableau.

SELECTED PROJECTS

- **awesome-industrial-datasets** — Curated 200+ starred GitHub repository for industrial datasets.
- **PlantPilot** — Conversational AI copilot for process plants using RAG and Gradio UI.
- **DigitalTwin** — Python library for plant monitoring (CPM, Kalman), deployed in Petrobras collaborations.

EDUCATION

Postdoctoral Researcher — Chemical Engineering, UFRGS 2019 – Present

Ph.D. — Chemical Engineering, UFRGS 2016 – 2019

Visiting Researcher — Artificial Intelligence Group, University of Kaiserslautern, Germany 2018

M.Sc. — Chemical Engineering, UFRGS 2015 – 2016

B.Sc. — Chemical Engineering, FURG 2010 – 2014

AWARDS

- Top Graduate, Chemical Engineering (2014)
- International Challenge Winner — Computer Vision for Scrap Inspection (2019)
- Best Industry 4.0 Application — National Prize (2021)